Day 1 started off very slow as we were waiting for other colleagues to join us. I last did in C++ in 24 years ago at University of Johannesburg as first year student and I wondered if I will remember.

The first day of the programme was both exciting and a little overwhelming because there was a lot of new information to take in. During the orientation, I was introduced to the course topics, and the expectations for the programme.
During orientation Bertie explained the beginning of C++ and how it has evolved of the decades. I was reintroduced to C++ via Blocks to C++. I loved the drag and drop as It was easier to create programs . As we dragged and dropped C++ code was being generated. It was easier to see how code works in C++. We created the “Hello World” program.

Frede had already introduced Git to us at Level 2 week. This was just a recap and we created a repository for our Level 3 C++ training work. I confidently say I can use basic Git now. I know how to change directory, add, commit and push the work.

Overall, the first day laid a strong foundation for the week ahead.

Day 2: My highlight for Day 2 was Ms V Booi, she is exceptional in explaining C++. She explained every concept clearly. She would first explain a concept by making a real life scenario, then explain same concept using C++ and lastly do a practical demonstration of the concept.

We learned the fundamentals of C++. It was great session, full of practical examples. As I learned about data types in C++, I began to appreciate how important it is to choose the right type of data for a program. At first, it seemed like a small detail, but through practice I realized how it affects the accuracy of the results. Learning how to read input from users and display output also helped me understand how programs communicate with people. Reflecting on this experience, I feel more confident in my ability to write basic programs and solve simple problems and gave me strong foundation in programming.

Day 3: Learning about loops and sentinel values helped me understand how programs can repeat tasks without writing the same code over and over again. At first, I found loops a bit confusing, especially in how I had to use them in programs. As we practiced in class, it became easier using them to solve different problems. I am comfortable using for loop, I still need more practice on the while and do-while loop. Especially the do while!

I also learned how sentinel values can be used to end a loop when the user is finished entering information. This helped me see how programs can handle situations where the number of inputs is not known at the start of the program.

Today has improved my understanding of Loops programming , I can create programs that repeat actions, process user input, and produce the correct results.

Day 4: Functions and arrays – My Nightmare at University. I dreaded this day because back in the day I struggled with these. Functions and arrays looked difficult compared to the topics I had learned before. Mr Tumelo made it easier to understand. It took me a while to get into the lesson because of my fears but now I confidently say give a program with function or arrays I will murder it. I have learned that functions help organize code into smaller, manageable sections, while arrays make it easier to store and work with multiple values. functions and arrays make programs more efficient and easier to manage..

Although I still found some parts challenging, I will keep on practising until I am to impact the knowledge to the next generation perfectly. Good day overall.

Day 5 Today was not a good day for me personally, very anxious.
We learned about Classes and Objects in C++. I struggled with the most today. I found it difficult to understand how classes and objects worked and what's they importance. I felt lost when trying to apply them in my programs and I constantly needed explanations from the facilitator and my peers. I had to read the same examples several times before they started to make sense. Although it was not easy, I began to understand how classes help organize code and how objects can be used as more real scenarios where given. Even though classes and objects are not a bit challenging to understand. I will continue to do small programs to enhance my my understanding.

Overall Reflection on the C++ Programme

Looking back on this past week, I can honestly say that I am 100% better at C++ than I was when I was a week ago. What initially seemed challenging, gradually became easier to understand through practice and guidance. I gained valuable knowledge in C++ programming in how to use data types, user input, decision-making, loops, functions, arrays, and object-oriented programming. Each lesson helped brought a little bit of confidence in programming and strengthened my problem-solving skills. The programme was not only about learning how to write code, but also about learning how to think logically and troubleshoot errors.

I would like to extend my sincere gratitude to **DHET and TUT** for providing this opportunity. The programme was well-organised, the facilities were top-notch, and the learning environment was conducive to growth and development. A special thank you goes to the facilitators, who were exceptional throughout the programme. Their knowledge, patience, and willingness to assist made a significant difference in learning this week.

I would also like to give a big shout-out to the catering team for the week. The food was consistently delicious and greatly appreciated, helping to keep us energized.

Overall, this was truly a great learning experience. I leave the programme with new skills, greater confidence, and a stronger interest in teaching programming. I am grateful for the opportunity and excited to continue building on the foundation that I have gained during this week.